quantum physics that have been proposed to date. While the different possibilities laid out in this book are the most significant historically, and they are all mostly still around in various forms (minus Wigner's consciousness-based proposal, which has been dismissed as needlessly speculative and vague, and in danger of collapsing into solipsism), many, many more have been proposed in the past thirty years. There are retrocausal interpretations, which suggest that subatomic particles can affect their own pasts, taking quantum nonlocality to an extreme. There are interpretations that try to get around Bell's theorem by altering the axioms of probability itself, though it's not clear if they can succeed. 't Hooft is developing his own interpretation of quantum theory that takes an unusual avenue through the obstacle course of weirdness set up by Bell's results. His theory is "superdeterministic," a local hidden-variables theory that has deep prearrangements between subatomic particles and experimental setups. Many physicists and philosophers reject this kind of approach out of hand as a sort of cosmological conspiracy theory that would preclude the very possibility of doing science. Yet 't Hooft believes that he can find a way to do this without sacrificing science itself, and he may not be wrong. Roger Penrose, one of the foremost mathematical physicists alive, believes that wave function collapse is real and that the Schrödinger equation must be modified, as in spontaneous-collapse theories. But rather than collapse being entirely random, he believes that it is caused by gravity, marrying general relativity and quantum physics in an unexpected and novel fashion. There are even interpretations that are a kind of hybrid of existing interpretations, like the many-interacting-worlds interpretation, which has features of both the pilot-wave and many-worlds interpretations.

There are also challenging issues in the interpretation of quantum field theory, the theory that combines quantum mechanics with special relativity and describes the intricate high-energy physics seen at particle accelerators. QFT shares some of the problems of regular quantum theory—the measurement problem and nonlocality are still there—but it also has some new and strange foundational problems of its own. Getting some of the existing interpretations of quantum theory to work with QFT, such as the pilot-wave interpretation, is an ongoing challenge. (Others, such as many-worlds, have no problem with QFT, which is arguably a mark in their favor.) And there are so, so many other ideas and open problems in